Supplementary Information: Residual-calibrated hedging

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Abstract

Supplementary Notes 1–2. Full study design: `EXPERIMENT_PLAN_SIDE.md` in the code release.

Supplementary Note 1: pre-specified hypotheses and decision rules

Each rule below was written into the study plan before the runs it concerns.

- **Study 1 (4 October 2026).** H1: committed-prefix minus time-aligned return > 0 at each of $d = 2, 4, 8$, pooled over cells above the 0.15 floor. Result: $p = 0.050$, not supported. H2: committed-prefix removes at least half of the time-aligned mismatch loss at $d = 4, 8$. Result: not supported (27–31%).
- **Hedging pilot (5 October 2026, before its runs).** Contrast $R(K = 16) - R(K = 1)$ at $d = 2, 4, 8$ on one model per cell. Rule: proceed if the 95% interval lies above 0 at two or more delays; fall back if it includes 0 at all three. Result: proceed ($d = 8$ and, marginally, $d = 4$).
- **Replication (5 October 2026, before its runs).** Over 24 models, $R(K = 16) - R(K = 1)$ at $d = 4$ and $d = 8$, one-sided, Holm-corrected. Secondary: $d = 2$, and $K = 4$ against $K = 1$. Result: supported at $d = 8$ (Holm $p = 0.0014$), not at $d = 4$ ($p = 0.19$).
- **Equal latency (5 October 2026, before its runs).** $R(K = 4, N = 512) - R(K = 1, N = 512)$ and $R(K = 16, N = 128) - R(K = 1, N = 512)$ at $d = 4, 8$, Holm over the four contrasts. Result: $K = 4$ supported at both delays (Holm $p = 0.035$, 0.0004); $K = 16$ at $N = 128$ worse at both.
- **Held-out method study (6 October 2026, before its runs).** Tuning of the hand-off and risk penalty strengths on the seed-0 models with separate episodes, by

a fixed rule (chosen: $\lambda_h = 0.25$, $\lambda_r = 0.5$; the hand-off penalty lowered tuning return at every strength). Primary contrasts on the 16 seed-1/seed-2 models at $d = 4, 8$, Holm over eight:

- C1: learned error model vs residual bank. Result: not supported (+0.001, +0.011).
- C2: hand-off penalty vs none. Result: not supported; the penalty lowered return (−0.026, −0.021).
- C3: hand-off vs risk penalty. Result: not supported (−0.011, +0.006).
- C4: full variant (learned model with hand-off penalty) vs committed prefix. Result: not supported (−0.001, +0.026).

Secondary: both hedges beat committed-prefix planning (+0.025/+0.037 for the bank, +0.025/+0.047 for the learned model, at $d = 4/8$).

Supplementary Note 2: compute

Studies 1–3 and the hedging studies: about 30 GPU-hours (Study 1 and 2 returns), 15 (pilot), 30 (replication), 8 (equal latency), 46 (diffusion), plus timing; no training. Recorded totals: [PENDING: per study].